

AI Search Framework v2: Decomposed All-SLM Pipeline with Feedback Loop

Study: Multi-LLM Baseline Roster & Feedback Loop Benchmark | **Scope:** Strictly Zero LLMs in Proposed Pipeline ($\leq 8B$) | **Evaluation:** N=80 within-subject paired trials with Blind LLM Judge (Claude-3.5-Sonnet)

Executive Summary

The v2 AI Search Framework investigation benchmarks a multi-stage all-SLM pipeline featuring a **bounded re-decomposition feedback loop** (max depth = 3) against a 5-model baseline roster across parameter scales: Llama-3.1-8B, Qwen-2.5-32B, Llama-3.1-70B, Qwen-2.5-72B, and Gemini-1.5-Pro.

Key Empirical Results (from results/v2_aggregated_results.json):

- Cost Savings vs 70B & Frontier:** Achieved an aggregate **0.584× cost ratio vs Llama-3.1-70B (41.6% savings)** and **0.111× vs Gemini-1.5-Pro (88.9% savings)**.
- Scale Crossover:** Economic crossover boundary established at $\approx$ **35B parameters**. Single models $\leq 32B$ incur lower invocation overhead than decomposition, but large & frontier models are substantially more expensive.
- Feedback Loop Precision (RQ5):** Re-decomposition reduced Graph Edit Distance (GED) from **3.57 $\rightarrow$ 1.50** (a **58.33% reduction in graph errors**).
- Independent LLM Judge:** In randomized blind judging, Qwen-2.5-72B captured 25.0% of wins, Gemini-1.5-Pro 18.75%, Llama-3.1-70B 15.0%, Llama-8B 15.0%, and SLM Pipeline v2 13.75%.

1. Empirical Results: Multi-Baseline Comparison

Baseline System	Param Scale	Cost Ratio (SLM / Baseline)	95% Cost CI	Parallel Speedup	Wall Latency Ratio
Llama-3.1-8B	8.0B	2.656×	[2.508, 2.805]	0.022×	1.48×
Qwen-2.5-32B	32.5B	1.178×	[1.112, 1.244]	0.021×	1.52×
Llama-3.1-70B (v1 match)	70.6B	0.584×	[0.551, 0.616]	0.021×	1.53×
Qwen-2.5-72B	72.7B	0.584×	[0.551, 0.616]	0.022×	1.51×
Gemini-1.5-Pro	Frontier	0.111×	[0.104, 0.117]	0.022×	1.49×

2. LLM-as-Judge Win-Rate Distribution (N=80)

Evaluator: c1aude-3-5-sonnet-20241022 under randomized blind candidate presentation:

System Identifier	Total Wins	Win-Rate (%)	Single-Domain Win-Rate	2-Domain Win-Rate	3+-Domain Win-Rate
Qwen-2.5-72B	20	25.00%	25.0%	23.3%	26.7%
Gemini-1.5-Pro	15	18.75%	25.0%	16.7%	16.7%
Llama-3.1-70B	12	15.00%	15.0%	16.7%	13.3%
Llama-3.1-8B	12	15.00%	10.0%	20.0%	13.3%
SLM Pipeline v2	11	13.75%	15.0%	10.0%	16.7%
Qwen-2.5-32B	10	12.50%	10.0%	13.3%	13.3%

3. Feedback Loop Effectiveness (RQ5)

- **Loop Firing Rate:** Re-decomposition triggered on **47.5%** of compound queries.
- **Mean Graph Edit Distance (GED):** Reduced from **3.57 (No Loop)** → **1.50 (With Loop)**.
- **Structural Accuracy Gain:** **58.33% reduction** in decomposition errors.

4. Direct Replication of v1 Baseline (Llama-3.1-70B)

Metric	v1 Historical	v2 Measured	Consistency Status
Cost Ratio vs 70B	0.596× (40.4% savings)	0.584× (41.6% savings)	Replicated ($ \Delta < 2\%$)

Reproducibility Citation: All v2 runs traceable to results/v2_eval_dev_master.jsonl, results/v2_records/, and results/v2_aggregated_results.json.